

The Transformer Block

DATA 443/643 · Week 3, note 3.3

After Princeton COS 484 L8–L9: positions, residual, MLP, encoder vs decoder.

Lecture plan

- 1 Attention is a set; language is ordered
- 2 Add the residual; do not replace x
- 3 The MLP holds most of the weights
- 4 Encoder two sublayers; decoder three

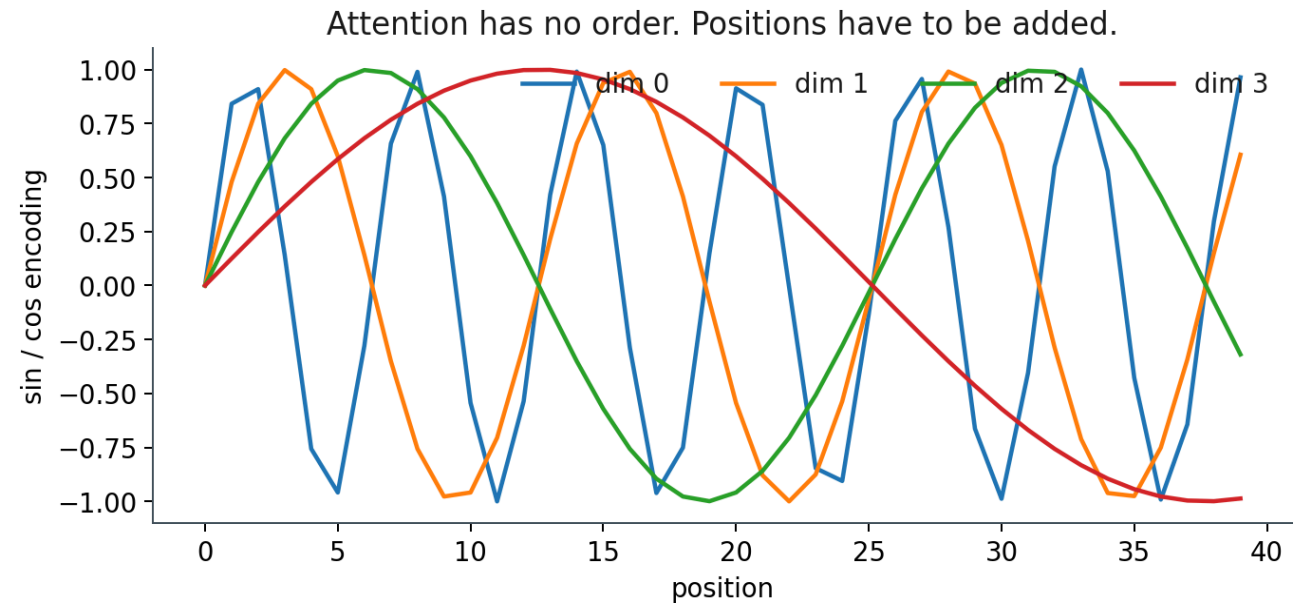
First time: what / why / formula

- What / why. One layer: (masked) self-attention, residual + LayerNorm, token-wise MLP, residual + LayerNorm. Positions added on the embeddings. Attention mixes across positions. The MLP mixes channels at one position. Residuals keep a highway so depth trains.
- Architecture. Decoder-only (GPT): causal mask. Encoder (BERT): bidirectional. Encoder–decoder: extra cross-attention.
- Formula. $X^{\ell+1} = X^{\ell} + \text{MLP}(X^{\ell} + \text{Attn}(X^{\ell}))$ (cartoon; norms omitted). Positions: $X_0 = E + P$.
- Tradeoffs. + Depth, parallelism, one block reused. – Quadratic attention; positions are extra parameters or sinusoids; encoder vs decoder is a method choice, not a vibe.

How. Add a positional vector to each token embedding. Pre-norm vs post-norm is a stability knob; we draw residual around sublayers.

Positions have to be added

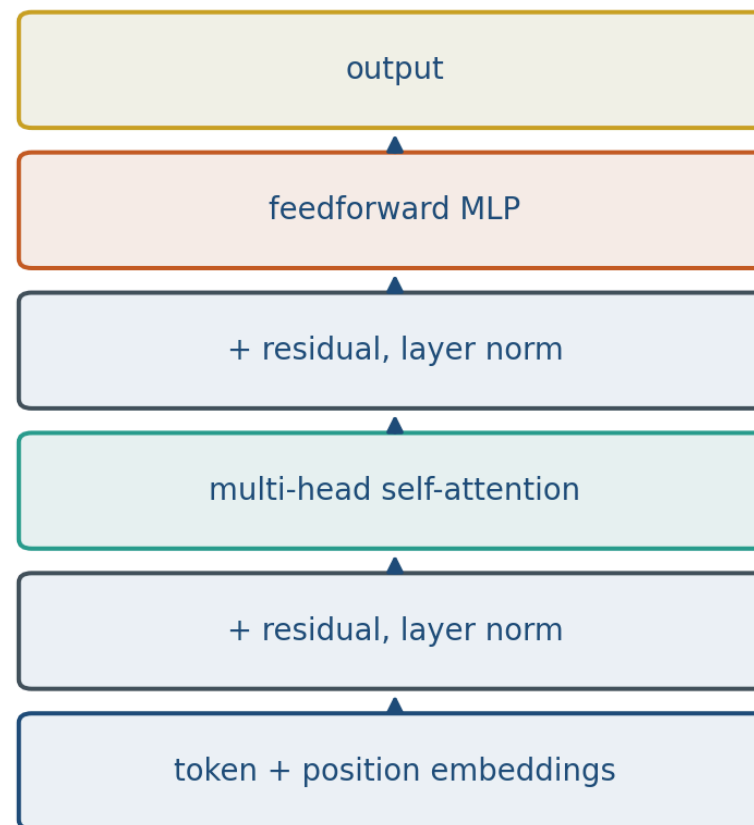
- Permute the tokens, attention permutes with them.
- Sinusoids or learned vectors, added to embeddings.
- Without them, dog bites man equals man bites dog.



One block

- Attention mixes across positions.
- The MLP mixes across channels at one position.
- Layer-norm one token at a time.

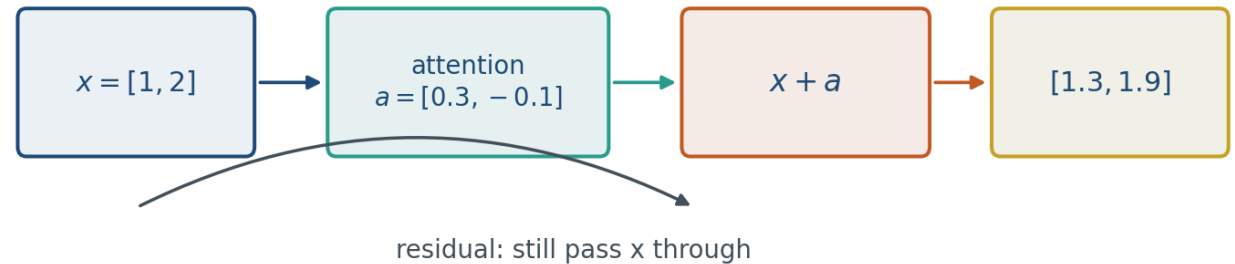
One transformer block



Residual: add, do not replace

- x becomes $x + \text{sublayer}(x)$.
- Cousin of the LSTM highway in Week 2.
- A dead head still passes x through.

Add, do not replace. Same idea as the LSTM highway.



Most weights are not attention

$$4d^2 \text{ (attention)} \quad \text{vs} \quad 8d^2 \text{ (MLP)}$$

W_Q, W_K, W_V, W_O versus d to $4d$ and back.

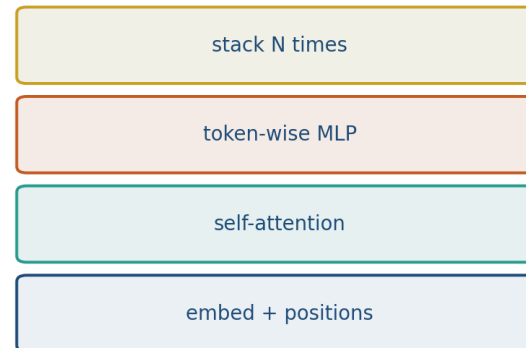
GPT-2 small: $N=12$, $d=768$.

Pre-norm is the modern default; post-norm is 2017.

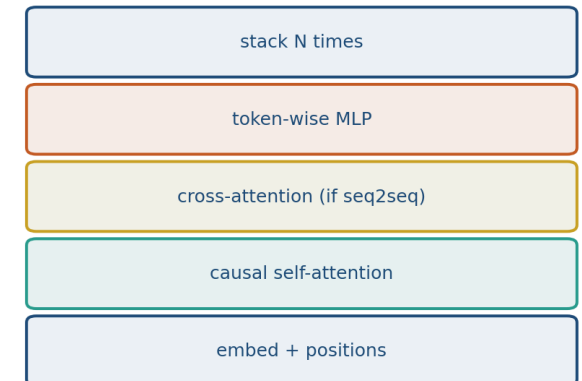
Encoder versus decoder

- BERT: encoder only, bidirectional.
- GPT: decoder only, no cross-attention.
- T5 / BART keep both. Captioning will too.

Encoder (BERT)



Decoder (GPT drops cross-attention)



What to take from this lecture

Order it

Positions are not optional for language.

Add it

Residual is x plus the sublayer, like a gated cell.

Count it

If you only plot attention heads, you missed the MLP.